

Classification on MNIST and FashionMNIST

1. Introduction

In this work, we evaluated the performance of **deep learning models** and **classical machine learning models** on the MNIST and FashionMNIST datasets. The objective was to study how model complexity, optimization strategies, and kernel-based methods affect classification accuracy. The datasets were divided into **70% training, 10% validation, and 20% testing splits**. Performance was evaluated primarily using **test classification accuracy**, and for SVM models, **training time** was also reported.

2. Deep Learning Models (ResNet-18 and ResNet-50)

2.1 Model Description

Two convolutional neural network architectures were used:

- **ResNet-18**: A relatively shallow residual network with 18 layers.
- **ResNet-50**: A deeper residual network with 50 layers and significantly more parameters.

Both models were trained **from scratch (no pretrained weights)**. Since MNIST and FashionMNIST consist of grayscale images of size 28×28, the images were resized to **224×224** and converted to **3 channels** to make them compatible with ResNet architectures.

2.2 Training Setup

- Optimizers used: **SGD** and **Adam**
- Learning rates: **0.001** and **0.0001**
- Batch sizes: **16** and **32**

- Number of epochs: **5–10 epochs**
- Loss function: **Cross-Entropy Loss**

The same training configuration was maintained across experiments to ensure a fair comparison.

2.3 Accuracy Analysis (CNN Models)

MNIST Dataset

Both ResNet-18 and ResNet-50 achieved **high classification accuracy** on MNIST even with a small number of epochs.

Key observations:

- **ResNet-18 consistently matched or outperformed ResNet-50.**
- Adam optimizer resulted in **faster convergence and higher accuracy** compared to SGD.
- Increasing model depth (ResNet-50) did **not significantly improve accuracy**, indicating that MNIST is a relatively simple dataset.
- Even with **5 epochs**, accuracies well above **90%** were achieved.

This shows that for simpler datasets like MNIST, **shallower architectures generalize better** and avoid unnecessary over-parameterization.

FashionMNIST Dataset

FashionMNIST is more complex than MNIST due to higher intra-class variability.

Observations:

- Accuracies were lower compared to MNIST but still competitive.
- Adam optimizer again outperformed SGD.

- ResNet-18 showed **better stability** than ResNet-50.
 - Increasing epochs improved accuracy, indicating FashionMNIST requires more learning capacity.
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2.4 Summary of CNN Results

- **Adam + learning rate 0.001** gave the best results.
 - ResNet-18 was more efficient and robust than ResNet-50.
 - Deep models converge very quickly on MNIST.
 - Model complexity should be matched to dataset complexity.
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3. Support Vector Machine (SVM) Models

3.1 Model Description

Support Vector Machines (SVMs) were trained using the following kernels:

- **Radial Basis Function (RBF)**
- **Polynomial (Poly)**

Since SVMs operate on fixed-length vectors, images were **flattened into 1D feature vectors** and normalized. Due to computational limitations, SVMs were trained on a **subset of 10,000 training samples**, which is standard practice for large datasets.

3.2 Training Setup

- Kernels: `rbf`, `poly`
- Regularization parameter: **C = 1.0, 10.0**

- Training time measured in **milliseconds (ms)**
 - Evaluation metric: **Test classification accuracy**
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3.3 Accuracy and Training Time Analysis (SVM Models)

MNIST Dataset

- **RBF kernel achieved the highest accuracy**, often comparable to deep learning models.
- Polynomial kernel performed slightly worse but required **less training time**.
- Training time for RBF kernel was significantly higher due to non-linear kernel computations.

FashionMNIST Dataset

- Overall accuracy was lower than MNIST.
 - RBF kernel still outperformed polynomial kernel.
 - Training time increased noticeably compared to MNIST, reflecting the higher complexity of FashionMNIST.
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3.4 Summary of SVM Results

- RBF kernel provides better accuracy but at the cost of **high computational time**.
 - Polynomial kernel is faster but less accurate.
 - SVMs do not scale well with dataset size compared to CNNs.
 - Deep learning models are more suitable for large-scale image classification tasks.
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4. Overall Comparison and Discussion

- **CNNs significantly outperform SVMs in scalability and training efficiency.**
 - SVMs are effective for smaller datasets but become computationally expensive for large image datasets.
 - ResNet-18 provides an excellent balance between accuracy and computational efficiency.
 - Increasing model depth does not always improve performance, especially on simpler datasets like MNIST.
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5. Conclusion

This study demonstrates that:

- Deep learning models, particularly ResNet-18, are highly effective for image classification tasks.
- Adam optimizer consistently provides better convergence and accuracy.
- SVMs with RBF kernels achieve strong performance but are limited by training time and scalability.
- Model selection should consider dataset complexity, computational cost, and desired accuracy.